1 Water

* Introduction:-

71% of earth surface is covered by water but from that 97% water in ocean is salty, 3% of fresh water is available.

Q. What are the sources of water?

i Surface - River, Sea, Ponds, Reservoirs.
ii Underground - wells, tubewells.
iii Rain - Purest form of water.

Q What are the types of impurities in water: mention methods to remove them (4M)

⇒ i) Suspended - (not completely soluble in water - particles) Impurities like soil, sand, organic waste industrial waste, mud present in water & size is greater than 1000Å.
Separate by filtration.

ii) Colloidal impurities:- (remain in continuous motion) Clay mud organic matter size is 10 to 1000Å colloidal particles. They can not removed by filtration
Can be removed ~~by filtration~~. By using Potash alum, $FeSO_4$, Sodium aluminate as a coagulant then sedimentation takes place & filtration.

iii) Biological impurities:-
Microscopic algae, fungi, bacteria, (aerobic, anaerobic pathogen etc) & other small size aquatic livings.
Removed by sterilization process chemicals - chlorine, bleaching powder, sod. hypochloride, ozone U.V. light is used.

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iii) (i) Dissolved impurities :-

Two types :-

a) Dissolved gases :- CO_2, Cl_2, H_2S, SO_x, NO_x

b) Dissolved salts :- when water percolate thr rock, soil some salts dissolved like bicarbonate, carbonate, sulphate, chlorites, nitrate of Ca, Mg, Fe, Al, Na, K

Removed by special softning procedure, chemical treatment, gases removed by warming, mechanical ~~deo~~ deaeration.

* Hardness of water :-

What is hard water ? or hardness of water ? 3M

⇒ i) Water contain various salts dissolved in it.

ii) Na, K is present ~~more~~ but not problematic.

iii) Ca, Mg is need to removed - $CaCl_2$, $MgCl_2$, $CaSO_4$, $MgSO_4$, $FeSO_4$, $Al_2(SO_4)_3$ etc.

iv) On the basis of salts it devided in to two types. —

a) Hard water	b) Soft water
do not formed lather easily with soap	formed lather easily
heavy ~~meata~~ metals like Ca^{++}, Mg^{++}, Fe^{++}, Mn^{++} & Al^{+3} dissolved in water in the form of carbonate, $(HCO_3)_2^-$, Cl_2, SO_3^-, NO_3^-	Na, K, Li dissolved in soft water (light metals)

* Causes of Hard water :-

Q Define Hard water ? Explain how rain water becomes hard.

or

Q What are the causes of hardness of water ?

1) Dissolution of minerals:-
Rain water flow & when it deposites on ground it percolate that time various salts dissolved in it.

2) Action of Dissolved CO_2
When rain water combine with CO_2 it gives soluble acidic gas.
$CaCO_3 + H_2O + CO_2 \longrightarrow Ca(HCO_3)_2$

3) Action of O_2
Oxygen slightly soluble in water - insoluble minerals to soluble & hardness causing salts.
$2FeS + 2H_2O + 7O_2 \longrightarrow 2FeSO_4 + 2H_2SO_4$

* Types of Hardness of Water:
Q What are the types of hardness of water? mention the methods to remove?
⇒ 1) Temporary hardness or carbonate / alkaline
2) Permanent or non-carbonate / non-alkaline

1) Temporary hardness of water - dissolve impurities
The hardness in water due to presence of bicarbonate & soluble carbonate of Ca, Mg, Fe, Al, Mn, salts:
$Ca(HCO_3)_2$, $Mg(HCO_3)_2$, $MgCO_3$, $FeCO_3$.
2) Soluble salts converted to insoluble salts by boiling and removed by filtration & clark's method ($Ca(OH)_2$)
3) Known as carbonate hardness or alkaline hardness.
i) $Mg(HCO_3)_2 \xrightarrow{\Delta} Mg(OH)_2 \downarrow + 2CO_2 \uparrow$
ii) $Ca(HCO_3)_2 \xrightarrow{\Delta} CaCO_3 \downarrow + CO_2 \uparrow + H_2O$
iii) $MgCO_3 + H_2O \xrightarrow{\Delta} Mg(OH)_2 \downarrow + CO_2 \uparrow$
iv) $MgCO_3 + H_2O \xrightarrow{\Delta} Mg(OH)_2 \downarrow + CO_2$

2) Permanent hardness / Non Carbonate
Due to dissolved impurities of heavy metals other than bicarbonates like Ca, Mg, Al, Fe, Mn etc.

Salts are $CaCl_2$ $MgCl_2$, $CaSO_4$, $MgSO_4$, $Ca(NO_3)_2$, $Mg(NO_3)_2$ etc

2) A: It can not be removed by boiling. need some chemical treatment.

3) Known as non-carbonate hardness or non-alkaline hardness.

4) Salts removed by chemical treatment or membrane filtration.

i) Soda lime addn 2) Sod. phosphate addn 3) Zeolite

4) Reverse osmosis. 5) Electrodialysis.

* Total Hardness :- Temp. Hardness + Permanent hardness.

* Degree of hardness :-

Hardness Producing Salt → It expressed in terms of $CaCO_3$ equivalence.

a) mol. wt. = 100.

b) Insoluble salt

$$\boxed{CaCO_3 = 40 + 12 + (16 \times 3) = 100}$$

$$CaCO_3 \text{ equivalent of any salt} = \text{wt. of salt present in (mg/lit)} \times \frac{\text{Eq. wt. of } CaCO_3\ (50)}{\text{Eq. wt. of salt.}}$$

$$CaCO_3 \text{ equivalent of any bivalent salt} = \text{wt. of salt present in (mg/lit)} \times \frac{\text{mol. wt. of } CaCO_3\ (100)}{\text{mol. wt. of salt.}}$$

* Units of Hardness :-

1) Milligrams Per liter (mg/lit)

2) Parts Per ~~mil~~ million (PPM) 10^6

$1\ PPM = 1\ mg/l = 0.1\ ^\circ Fr = 0.070\ ^\circ Cl$

French Clark

$$eq\ wt. = \frac{m.\ mass}{charge/acidity/basicity}$$

H = 1, Ca = 40, Mg = 24, S = 32, O = 16, C = 12, Fe = 56, Na = 23, Al = 27, Cl = 35.5

Numericals:

1 Water sample contain 16.2 mg of $Ca(HCO_3)_2$, 36 mg of $MgSO_4$, 27.2 mg $CaSO_4$, 2 mg of $MgCO_3$. Find the temp. per. & total hardness of water.

eq. wt. $Ca(HCO_3)_2 = 81$, $MgSO_4 = 60$, $CaSO_4 = 68$, $MgCO_3 = 42$

	Chemicals	amount in mg/lit	mol. wt./ eq. wt.	$CaCO_3$ equivalent in mg/lit	Types of Hardness
1	$Ca(HCO_3)_2$	16.2	81	$\frac{16.2 \times 50}{81} = 10$	Temp
2	$MgSO_4$	36	60	$\frac{36 \times 50}{60} = 30$	per
3	$CaSO_4$	27.2	68	$\frac{27.2 \times 50}{68} = 20$	Per
4	$MgCO_3$	2	42	$\frac{2 \times 50}{42} = 2.38$	Temp.

$$CaCO_3 \text{ equivalent of any salt} = \text{wt. of salt present in mg/lit} \times \frac{\text{Eq. wt. of } CaSO_4 (50)}{\text{Eq. wt. of salt}}$$

Temporary hardness = $Ca(HCO_3)_2 + MgCO_3$ (hardness)
= 10 + 2.38
= 12.38 PPM $CaCO_3$ equivalent.

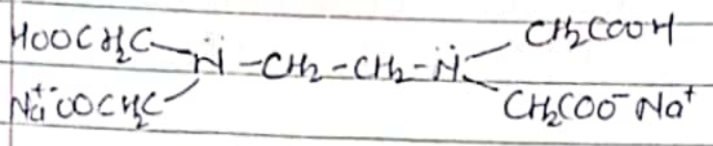
Permanent hardness = $MgSO_4 + CaSO_4$ (hardness)
= 30 + 20 = 50 ppm $CaCO_3$ eq.

Total Hardness = Temp. Hardness + Per. Hardness
= 12.38 + 50
= 62.38 PPM $CaCO_3$ equivalent.

* Determination of hardness of water by EDTA method.

Q Explain EDTA method for determination of hardness of water. 6M 3M.

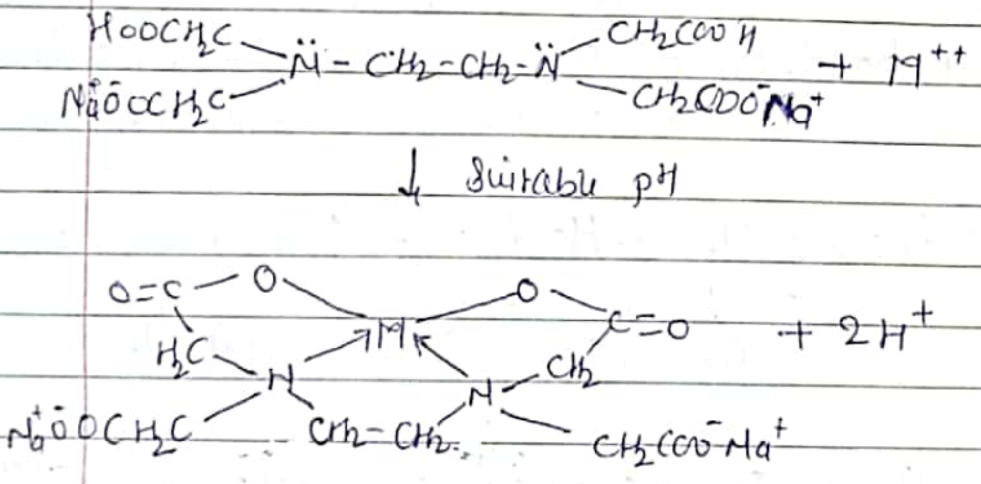
EDTA method is also called complexometric titration used to determined hardness of water. - Complexing agent - EDTA - Ethylene diamine tetraacetic acid.
" Total hardness of water sample is found accurately by titration of the water sample with EDTA".



Disodium EDTA.

Q5 Draw metal EDTA complex & given chem reacn ?

1) Highly sensitive toward heavy metals ion in water
2) Called Hexadentate. - (with 4 carboxylic gr + 2 N lone pair e^-)
3) Suitable pH is reqrd to complex reacn.
4) If H^+ ions formed pH get decreases.
5) Reacn betn EDTA & heavy metal - cyclic co-ordination complex - hence called complexometric titration.



Metal Na EDTA complex. ($M^{++} = Ca^{+2}, Mg^{++}, Fe^{++}$, etc)

6) EBT - Eriochrome Black T indicator is used.

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Principle :-

1) Hardness causing ions like Ca^{+2} & Mg^{+2} present in hard water which make stable, soluble complex with disodium EDTA is called complexometric titration.

2) EBT - is used at pH 10 Ammonia buffer is used to maintain pH. :- ($NH_4OH + NH_4Cl$)
(EBT added to water at pH 10 & combn with Ca^{+2} & Mg^{+2} ions gives weak soluble complex wine red colour)

3) After titration unstable complex (wine red) ~~titrate~~ with EDTA - gives colourless complex. When all metals (ions) react with EDTA regeneration of blue colour of EBT occures.

4) end pt. wine red to blue

Reacn

1) $M^{+2} + EBT \longrightarrow M\text{-}EBT + 2H^+$
wine red

2) $M^{+2} + EDTA \xrightarrow{PH\ 10} M\text{-}EDTA + 2H^+$
colourless

3 $M\text{-}EBT + EDTA \longrightarrow M\,EDTA + EBT$
(wine red) (colorless) colorless Blue

1) Ca^{+2} or Mg^{+2} + EBT $\xrightarrow{PH\ 10}$ Ca-EBT
Mg-EBT
wine red - unstable complex

2) Ca-EBT
Mg-EBT + EDTA $\xrightarrow{PH\ 10}$ Ca-EDTA
Mg-EDTA + EBT
colourless Blue colour
stable complex

Std $MgSO_4$ (25ml) + 15ml Buffer + 4-5 drops EBT → wine red + EDTA → blue
Burette - EDTA
Conical flask - $MgSO_4$

* Procedure :-

Step. I - Std. of EDTA Solution.

1) Prepared standard soln of $MgSO_4$.
2) Filled Burette with EDTA & pipette out 25 ml Std - $MgSO_4$ soln in conical flask.
3) Add 15 ml of buffer soln at pH 10 & 4-5 drops of EBT indicator.
4) Titrate wine red colour soln with EDTA till the colour changes blue.
5) Note down that reading X ml.

Step II - Hardness of water sample :-

1) Take V ml (50ml) water sample in conical flask.
2) add 15 ml of buffer at pH 10 & 4-5 drops of EBT
3) Titrate wine red to mix. against EDTA. till the colour changes to end pt. blue.
4) Note down reading Y ml.

* Calculations :-

1 Standardisation of EDTA.

molarity of $MgSO_4$ is M_1 molar, Cal. Molarity of EDTA (Disodium) using volm of $MgSO_4$ soln volm of di Na EDTA.

$$M_1V_1 = M_2V_2$$
$$MgSO_4 = EDTA$$

2 Total Hardness of water sample.

1M of EDTA = 1M Ca^{+2} or Mg^{+2} = 1M $CaCO_3$ equivalent

∴ 1000 ml of 1M EDTA = 100 gm of $CaCO_3$

∴ Total Hardness of water = $\frac{Y}{V} \times M_2 \times 100 \times 1000$ ppm of $CaCO_3$ equivalent

V - Vol^m of water sample
Y - Vol^m of disodium EDTA
M_2 - molarity of disodium EDTA

* ~~Total~~ Hardness of water
In this method (EDTA) we get temporary & permanent hardness.
2) After boiling temp. Hardness removed only permanent hardness is left.

$$\text{Permanent Hardness} = \frac{Z}{V} \times M_2 \times 100 \times 1000 \text{ PPM}$$

$$\text{Total Hardness} = \text{Temp Hardness} + \text{Per. Hardness}$$

$$\boxed{\text{Tem Hardness} = \text{Total Hardness} - \text{Per. Hardness}}$$

eg - 50 ml of a water sample reqrd 12.7 ml of 0.02M EDTA during titration. cal. total hardness of water.

V = vol^m of water = 50 ml.
Y = vol^m EDTA = 12.7 ml.
M_2 = molarity of EDTA = 0.02M

$$\text{Total hardness} = \frac{Y}{V} \times M_2 \times 100 \times 1000 \text{ ppm } CaCO_3 \text{ eq.}$$

$$= \frac{12.7}{50} \times 0.02 \times 100 \times 1000$$

$$= 508 \text{ PPM } CaCO_3 \text{ equivalent.}$$

* Alkalinity of water sample :-
Q How alkalinity of water sample determined?
Q Explain it with procedure, formula & table of determination.

⇒ Alkalinity is due to presence of HCO_3^-, CO_3^{2-}, OH^- of Na, K, Ca, & Mg
[Alkalinity is ability of water to neutralize acid]

2

It Classify as :-
1) Bicarbonate alkalinity
2) Carbonate alkalinity
3) Hydroxide alkalinity
Strength $OH^- > CO_3^{-2} > HCO_3^-$
(Combn of OH^- & HCO_3^- not possible gives - carbonate

* Theory :-
1 When an alkaline water is titrated with acid (Strong) All OH^- get neutralised, then CO_3^{-2} ions are half neutralised to HCO_3^- upto phenolphthalein end pt.
2) pH get decreases till 8.7 at this stage soln becomes pink colour to colourless
3 Continue addn of acid during titration. all HCO_3^- get neutralised & complete the reacn indicate by methyl orange. (colour changes yellow to orange) at pH 4.
Reacn

1) $OH^- + H^+ \longrightarrow H_2O$
2) $CO_3^{-2} + H^+ \longrightarrow HCO_3^-$ } P ; Pink → colourless
3) $HCO_3^- + H^+ \longrightarrow H_2O + CO_2$ } M - yellow to orange

Procedure :-
1) Take V ml (25 ml) of the alkaline water sample in conical flask & add two drops of phenolpthalene indicator in it (gives pink colour)
2) Titrate that sample with strong acid from burette till pink colour changes to colourless. let the burette reading V_1 ml.
3) Add methyl orange indicator into same sample & continue the titration till the yellow colour of mix. changes to orange Note down the burette reading as V_2 ml

25 ml water + Phenolphthalene → Pink colour $\xrightarrow[\text{Titration}]{HCl}$ colourless = V_1 ml

↓ + methyl orange

Orange colour $\xleftarrow[V_2\,ml]{HCl}$ Yellow colour

Calculations:—

$$(0\,ml)\ \xrightarrow[\substack{\text{Phenolphthalene}\\ \text{end pt.}}]{OH^-,\ \frac{1}{2}CO_3^{-2}}\ (V_1\,ml)\ \xrightarrow[\substack{\text{methyl orange}\\ \text{end pt.}}]{\frac{1}{2}CO_3^{-2},\ HCO_3^-}\ (V_2\,ml)$$

concentration of acid

$$P = \text{phenolphthalene alkalinity} = \frac{V_1 \times 2 \times 50 \times 1000}{V} \text{ ppm } CaCO_3 \text{ equivalent}$$

V — water sample

$$M = \text{methyl orange alkalinity} = \frac{V_2 \times 2 \times 50 \times 1000}{V} \text{ ppm } CaCO_3 \text{ equivalent}$$

* **Cal. types of amount of alkalinity:**

The types & amount of OH^-, CO_3^{2-} & HCO_3^- alkalinity can be cal. by eq^n P & M.

$M = CO_3^{2-} + HCO_3^- + OH^-$, $2P = OH^- + CO_3^{-2}$

Sr No	alkalinity	quantity of OH^-	quantity of CO_3^{-2}	quantity of HCO_3^-
1	$P = 0$	0	0	M
2	$P = M$	P or M	0	0
3	$P = \frac{1}{2}M$ (carbonate 2 times)	0	2P or M	0
4	$P > \frac{1}{2}M$ ($2P > M$)	$(2P - M)$	$2(M - P)$	0
5	$P < \frac{1}{2}M$ ($2P < M$)	0	2P	M − 2P

$CO_3^{2-} + H^+ \rightarrow HCO_3^-$ P

$HCO_3^- + H^+ \rightarrow H_2CO_3$

Q. Cal. 50 ml of alkaline water sample req^d 5.2 ml of N/50 HCl upto phenolphthalene end pt. & 15.4 ml for complete neutralisation. Find the types & amount of alkalinity in water sample.

⇒ $V_1 = 5.2$ ml

$V_2 = 15.4$ ml

$V = 50$,

$Z = N/50 = \frac{1}{50} = 0.02N$

(N - Normality = solute in 1 lit

Phenolpthaleine alkalinity(P) = $\frac{V_1}{V} \times 2 \times 50 \times 1000$ (PPm)

$= \frac{5.2}{50} \times 0.02 \times 50 \times 1000$

$= 104$ PPm

Methyl orange alkalinity (M) = $\frac{V_2}{V} \times 2 \times 50 \times 1000$ PPm

$= \frac{15.4}{50} \times 0.02 \times 50 \times 1000$

$= 308$ PPm

Relⁿ betⁿ. P & M $\quad$ $P < \frac{1}{2}M$

CO_3^{-2} & HCO_3^- are present

$\therefore CO_3^{-2}$ alkalinity $= 2P = 104 \times 2 = 208$ PPM

HCO_3^- alkalinity $= M - 2P = 308 - 208$

$= 100$ PPM

hence CO_3^{-2} alkalinity present = 208 PPM

HCO_3^- —— 〃 —— = 100 PPM

* Effects of Hard water on Boiler:-

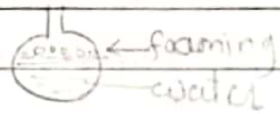
Q What are the hardness std of water for boiler? which problems arises due to untreated water?

⇒ water is used for steam generation in boiler this water should be free from impurities. In avoides the problem inside the ~~stea~~ boiler

Types of Boiler	hardness in feed water
1) Low pressure	25 - 50 PPM of $CaCO_3$ eq.
2) medium pressure	10 - 25 PPM —〃—
3) High pressure	0 - 10 PPM —〃—

alkaline solution to neutralise the product.
2) Boiler water should not contain salts.
3) Mg salts can be removed by zeolite process.

* Priming & Foaming:-
1) Priming (carry over)
"wet steam formation is called priming"
* Causes of priming:-
i) High level of boiler- feed water.
2) High steam velocities
3) Faulty boiler desine
4) presence of large amount of salts.
* Prevention of priming:-
i) avoiding rapid change in steaming rate.
2) maintained low water level.
3) Softning & filtration of boiler feed water

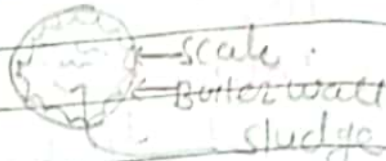
Foaming:-
Formation of continuous foam (bubbles) on the surface of water.
* Causes:-


Presence of sub^n like oils, soaps
* Prevention:-
i) Adding antifoaming chemicals likes castor oil.
2) Removing oil impurities by adding comp like Na-aluminate.
(Priming & foaming usually occures together)
* Disadvantages of priming & foaming:-
1) Dissolved salts deposited on blades & reduces efficiency.
2) Decreases life of machin.
3) actul hight of water column can not judge so maintainance of the boiler difficult.

* Sludge & Scale:-

Q Define sludge. Give the causes disadvantages

⇒ "The slimy loose deposits of ppt. salts in boiler. Salts deposites at the bends & valves affecting free flow of water.



* Causes:-

The subⁿ which have greater solubility in hot water than in cold water eg. $MgCl_2$, $CaCl_2$, $MgCO_3$, $MgSO_4$ etc.

* Disadvantages:-

1) sludge is poor conductor of heat so waste a portion of heat.
2) Decreases the efficiency of boiler.
3) Reduces water flow rate in boiler.

* Prevention:-

1) By using soft water.
2) By making Blowdown operation.

* Scales

Q What are causes, adv. & disadvantages of scale..

"The hard & strong coating formed inside boiler tube by chemical reacⁿ which is bad conductor of heat."

* causes (formation)

1) Deposition of bicarbonates:

At high temp bicarbonate decomposes into sticky water insoluble materials

$$Ca(HCO_3)_2 \xrightarrow{\Delta} CaCO_3\downarrow + H_2O + CO_2\uparrow$$

$$Mg(HCO_3)_2 \xrightarrow{\Delta} \underset{\text{scale}}{Mg(OH)_2\downarrow} + 2CO_2\uparrow$$

2) Hydrolysis of magnesium salts.

At high temp. Mg salts hydrolised & formed sticky magnesium hydroxide ~~salt~~ solid ppt.

$MgCl_2 + 2H_2O \rightarrow Mg(OH)_2 \downarrow + 2HCl$

$MgSO_4 + 2H_2O \rightarrow Mg(OH)_2 \downarrow + H_2SO_4$

3 Presence of silica.

Silica react with soluble salts of Ca & Mg forming ~~CaSO~~ $CaSiO_3$ or $MgSiO_3$ as firmly adhering materials.

4. Decreased solubility of $CaSO_4$.

Some salts presence in hard water which are insoluble in superheated water eg $CaSO_4$ & $CaCO_3$ will formed ppt. as hard scale formation

* Disadvantages:-

1) Wastage of fuel:-

Scales are bad conductor of heat so rate of heat transfer takes place from boiler to inside water is decreases. if ~~gr~~ Higher thickness of scale, greater is the wastage of fuel.

2) Lowering of boiler safety:-

Due to scale overheating takes place & boiler makes the boiler material softer & weaker. so it can not bear the pressure of the steam.

3 Decreases in efficiency:-

When scale deposite ~~the~~ in valves, condenser of the boiler & choke them partially. this decreases efficiency of the boiler.

4 Danger of explosion:-

Due to scale overheating takes place & cracks get developed. when water comes suddenly in contact with the overheated boiler metal. this causes large amount of steam formation & high pressure. Due to high pressure softer boiler metal may get burst.

* Removal of Scales:-

1) Use of scraper or wire brush to remove scale
2) By using suitable chemicals like EDTA. If scale they are hard or react with EDTA & formed soluble complex.
3) Thermal shocks teq. in this method empty boiler is heated & cooled by cold water suddenly. Scale can be made loose & removed easily.
4 Thick scale can be removed by hammer.

* Prevention:-

1 Use of softened water.
2 By certain chemicals to boiler feed water such treatment is called External treatment.
3 Adding sod. phalluminate which can trap the scale formation.
4 Frequent blow down operation. to remove the sludges & ppt.

5)

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* Water treatment :-
1) External treatment
2) Internal treatment

1) External treatments :- zeolite, ion exchange, RO
Treatment in which scale forming & corrosive impurities are removed from water before adding in boiler

2) Internal treatments :-
various sub^n are added to boiler feed water to prevent scale formation & boiler corrosion. calgon

External treatments :-
1) zeolite / Permutit method
2) Demineralization / Deionisation

* Softening of water by zeolite process (Cation exchange) (Permutit)

Q What are zeolites? Explain zeolite process of softning of water. Give regeneration reac^n, advantages & disadvantages of the process.

⇒ zeolite - zein - boiling, Lithos - stone
[$Na_2O \cdot Al_2O_3 \cdot xSiO_2 \cdot yH_2O$ ie Na_2Ze
$x = 2$ to 10 [(Na_x^+ [$(AlO_2)_x \cdot (SiO_2)_y \cdot 2H_2O$]
$y = 2$ to 6

* zeolite - hydrated sodium alumino silicate.
* Each Si atom is bonded to four O atom & each Al atom is bonded to four O atom tetrahedrally.

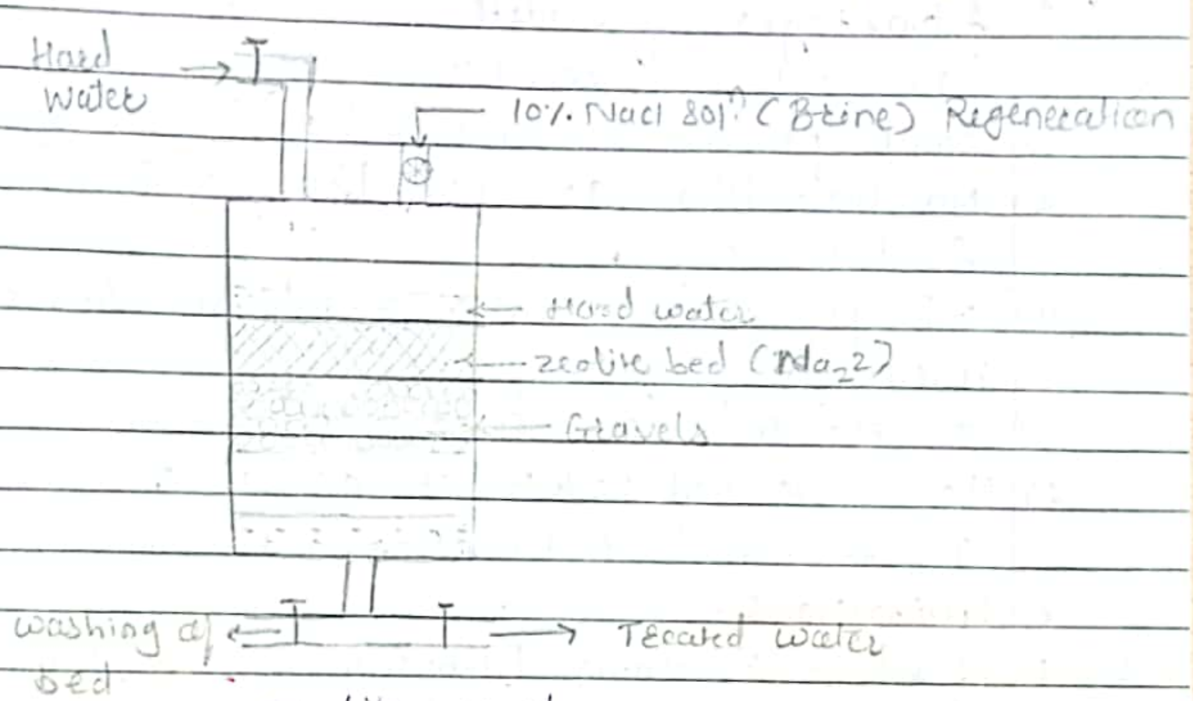
* Principle :-
Sodium zeolite has the property of capturing the heavy metal ions from water & in exchange release the Na^+. Due to removal of heavy metal ions (Ca^{++}, Mg^{++} etc) from water, the hard water is converted in-to soft water by zeolite

Types of zeolite – Natural (nonporus, green sand by wa
Synthetic treatment with caustic soda – nonpo. more durable)

Prepared by Δ together China clay, feldspar & soda. then cooling & granulation. They can prepared by Δ solⁿ sod. silicate, Al-sulphate, & sod. aluminate – porus & gel structure.

Process :-

1) zeolites are crystalline, ~~microscopi~~ porus powder of sodium zeolite placed over the perforated plate



Zeolite Exchanger

2 There are two inlets one is for hard water & second inlet is for addition of 10% NaCl solⁿ at the bottom one outlet is there to collect soft water.

4 When ~~zeolite~~ Hard water percolate thʳ zeolite bed the hardness causing Ca^{++}, Mg^{++} cation retained in the bed & Na released in the water in exchange.

5 Outgoing water contain Na salts.

$$Na_2Ze + Mg^{+2} \longrightarrow MgZe + 2Na^+$$

$$Na_2Ze + Ca^{+2} \longrightarrow CaZe + 2Na^+$$

* Regeneration of zeolite :- $Na_2Ze + Ca(HCO_3)_2 \rightarrow CaZe + 2NaHCO_3$
$Na_2Ze + CaCl_2 \rightarrow CaZe + 2NaCl$

when hard water passed thʳ zeolite bed exchange of ions takes place, & Na from zeolite bed

(1) Amount of NaCl in mg = lit of NaCl × g of NaCl × 1000

(2) Amount of NaCl in term of $CaCO_3$ = $\frac{\text{mg of NaCl} \times 50}{58.5}$ — Eq. wt of NaCl

(3) Lit of Hard water = $\frac{\text{Amount of NaCl in term of } CaCO_3 \text{ eq.}}{\text{Hardness of water}}$

exchange with Mg^{+2} & Ca^{+2} zeolite.

2 It need regeneration by passing 10% NaCl sol^n.

$$Mg\,Ze + 2NaCl \longrightarrow Na_2Ze + MgCl_2$$

$$Ca\,Ze + 2NaCl \longrightarrow Na_2Ze + CaCl_2$$

* Advantages of zeolite :—

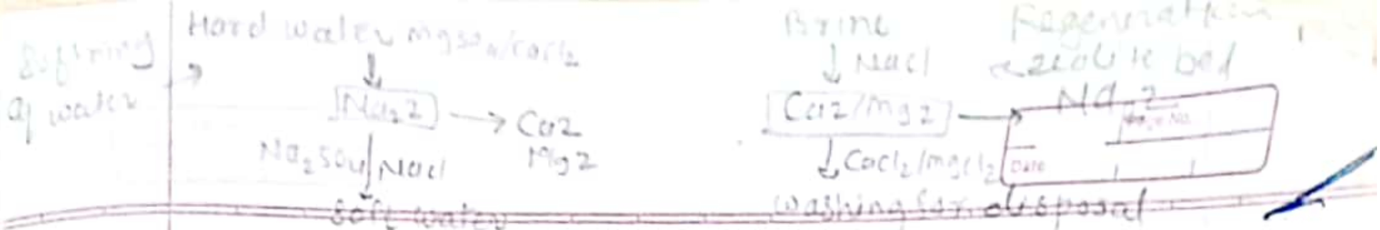
1) Low cost for operation.
2) Easy process to operate & Reqrd less space.
3) No impurities are present so no formation of sludge.
4) This process is atomatically adjust with Hard water itself.
5) It required less time for softning.
6) the zeolite bed is whether active or exhausted can be tested with soap. 7) less maintanance & skills

* Limitations :—

1 If water containing turbidity, suspended matter, colloider impurity should not be directly fed to the zeolite softner, because pores of zeolite bed will be clogged.
2) If water contain excess of acidity or basisity it destroy zeolite bed so pH should be 7
3.) Mn^{+2} & Fe^{+2} coloured ions destroyed the zeolite bed. So not regenerate with brine.
4 This method is useful only for small scale.
5 Its output water has no impurities (zero hardness) but contain equivalent quantity of Na salts.
6) only Ca^{+2} & Mg^{+2} ions are replace not HCO_3^-, CO_3^{2-}

* Applications of zeolite :—

1) Used to remove hardness causing ions from water.
2) Used to removed toxic metal ions & dyes cations from polluted water
(3 valuable metals can be recovered from industrial



waste water. eg. Ni, Cr, lead, Cu, Ag, etc.

* Cal.

1 A zeolite softner was completely exhausted & was regenerated by passing 100 lit. of NaCl containing 150 gm/lit of NaCl. How many lit. of sample of water containing hardness 600 ppm can be softened by this softner.

⇒ Step I - ~~mg of NaCl~~ = cal. of total quantity of NaCl

100 lit. of NaCl — 150 gm/lit. of NaCl.

Amount of NaCl = gm/lit x lit. of NaCl x 1000

$= 150 \times 100 \times 1000 = 1.5 \times 10^7$ mg NaCl

Step II Quantity of NaCl in term of $CaCO_3$.

58.5 mg NaCl = 50 gm $CaCO_3$ = $\frac{\text{amount} \times 50}{58.5} = \frac{1.5 \times 10^7 \times 50}{58.5} = 1.28 \times 10^7$ mg $CaCO_3$

1.5×10^7 = ?.

Step III Hardness x lit. of water = Quantity of NaCl 1.28×10^7 mg ~~to~~ $CaCO_3$

$600 \times$ Lit. of water $= 1.28 \times 10^7$

$= \frac{1.28 \times 10^7}{600} = 21333.3$ lit

Q. A zeolite bed exhausted by softning 4000 lit of water req^rd 10 lit. of 15% NaCl sol^n for regeneration. cal. the hardness of water sample.

* Step cal. total quantity of NaCl —

10 lit. of 15% NaCl is req^rd for regeneration.

15% means 15 gm NaCl in 100 ml water.

1000 ml sol^n contain 150 gm NaCl

10 lit — " — 1500 gm NaCl

* Step Cal. of $CaCO_3$ equivalence :-

NaCl added is cal^n by equivalent amount Ca^{+2}/Mg^{+2} which is cal $CaCO_3$ eq.

23+35.5

58.5 gm NaCl = 50 gm $CaCO_3$

1500 gm NaCl = x

$x = 1500 \times \frac{50}{58.5} = 1282.051$ gm $CaCO_3$ equivalent.

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Step 3 Cal. of hardness of water sample.
4000 lit. of water contain 1282.051 gm $CaCO_3$ equivalent hardness

1 lit = ? x

$$x = \frac{1282.051}{4000} = 0.3205 \text{ gm } CaCO_3$$

320.5 mg/lit. = 320.5 ppm $CaCO_3$ eq.

3 Zeolite softner was completly ~~ext~~ exhausted & was regenerated by passing 100 lit. of NaCl containing 120 gm/lit. of NaCl. How many lit. of sample of water of hardness 500 ppm can be softnen by this softner?

$\Rightarrow$ Amount of NaCl = gm/lit × lit. of NaCl × 1000
= 120 × 100 × 100
= 1.2×10^7

Amount of NaCl in $CaCO_3$ equivalent

$$= \frac{\text{amount} \times 50}{58.5} = \frac{1.2 \times 10^7 \times 50}{58.5}$$

$= 10.256 \times 10^6$ ppm of $CaCO_3$

3. Vol^n of water sample
Hardness × lit. of water = 10.256×10^6
500 × lit of water = 10.256×10^6

$$\text{lit. of water} = \frac{10.256 \times 10^6}{500}$$

= 20512.8 lit.

Q Find the hardness of water sample from the given data, A zeolite bed get exhausted on softning 2400 lit of water & reqd 10 lit. of 8% NaCl for regeneration

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$CaCl_2 \longrightarrow Ca^{++} + 2Cl$

$MgSO_4 \longrightarrow Mg^{++} + SO_4$

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* Water treatment by Ions Exchange Method (Deionisation / Demineralisation)

Q Describe ion 'exchange' method for Softning of hard water. 5 M, 6 M.

= Principle :-

Ion-exchange resins are insoluble, cross-linked long chain organic polymers with microporous structure, & functional gr. attached to the chain is responsible for ion-exchange properties.

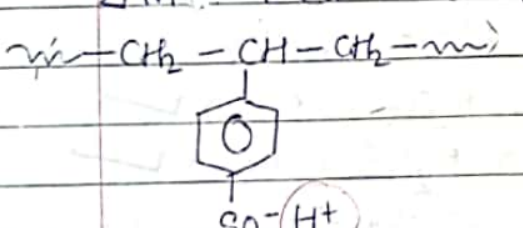
There are two types of resins -

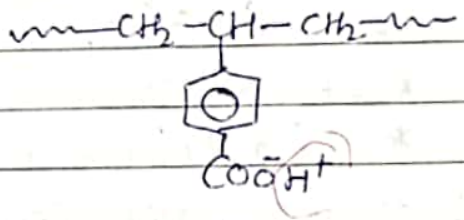
1) Cation exchange resins (RH^+)
2) Anion exchange resins ($R'OH^-$)

1 Cation exchange resins :

are mainly styrene-divinyl benzene copolymer, having carboxylated / sulphonated contain loosely held H^+ ion which can exchange easily with cation from water.

$$2RH + Ca^{+2} \longrightarrow R_2Ca + 2H^+$$

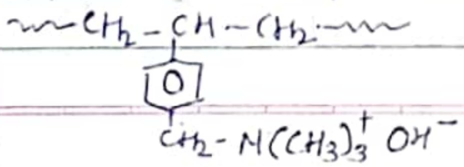




2 Anion exchanger (R-OH)

styrene-divinyl benzene (or amine formaldehyde) co-polymer which contain amino or quaternary ammonium gr. After treatment the resin with NaOH it exchange loosely held OH^- ion.

$$2R(OH) + SO_4^{-2} \longrightarrow R_2'SO_4 + 2OH^-$$



* Principle :-
When water containing dissolved salts is passed th^r cation exchanger all cation get exchanged with H^+ ions from resin. then water passed th^r anion exchanger all anions are exchange with OH^- ions from resins. so it become free from cation & anion i.e demineralised.

* Process :-

1) When water passed th^r cation exchanger 1st it removed all cation i.e Na^+, Mg^{+2}, Ca^{+2}, from water & H^+ ions released from this exchanger so water released from cation exchanger is acidic.

Rec^n :- ($2RH^+ + Ca^{+2} \longrightarrow R_2Ca^{+2} + 2H^+$)

eg - $Ca(HCO_3)_2 + 2RH \longrightarrow R_2Ca + 2H_2CO_3$

$MgCl_2 + 2RH \longrightarrow R_2Mg + 2HCl$

$Ca(NO_3)_2 + 2RH \longrightarrow R_2Ca + 2HNO_3$

2 This acidic water passed th^r anion exchanger all anions like, Cl^-, NO_3^- CO_3^{-2} from water & OH^- ions from exchanger are released.

eg - (R-$(OH) + Cl^- \longrightarrow R'Cl + OH^-$)

* eg - R'-$(OH) + HCl \longrightarrow R'Cl + H_2O$

* $2R'(OH) + H_2SO_4 \longrightarrow R'_2SO_4 + 2H_2O$

3 Thus water coming from exchanger is free from cation & anion. i.e ion free is known as deionised.

Regeneration :-

* i The exhausted cation exchanger is regenerated by washing with dil. HCl sol^n

$R_2Ca + 2HCl \longrightarrow 2RH + CaCl_2$

~~Rec~~ $R_2Mg + 2HCl \longrightarrow 2RH + MgCl_2$

[All H^+ ions replaced with Ca^{+2} or Mg^{+2}, then cation exchanger bed is said exhausted]

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2. like that all OH^- ions from anion exchanger get replace with Cl^- or SO_4^{-2} etc. bcz get exhausted.

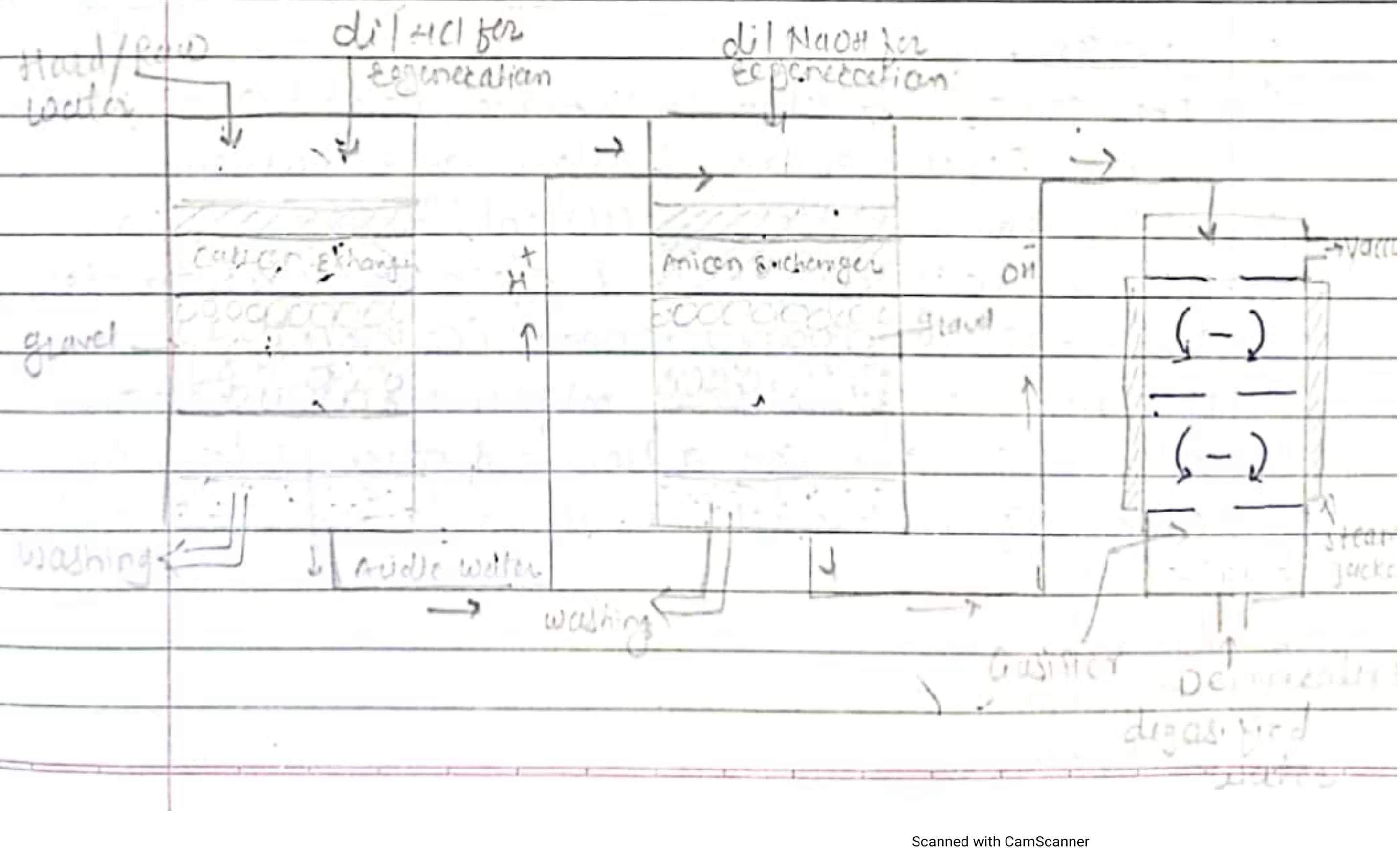
$$R'_2SO_4 + 2(NaOH) \longrightarrow 2R'(OH) + Na_2SO_4$$
$$\underset{\text{Exhausted}}{R'Cl} + \underset{\text{dil. base}}{NaOH} \longrightarrow \underset{\text{Regeneration}}{R'(OH)} + \underset{\text{washings}}{NaCl}$$

* **Advantages:-**

1) This process is useful for highly acidic or basic (alkaline) water.
2) water obtained from deionised has very low hardness 0-2 ppm. ie no salt
3) easy to operate.
4) Required less space. 5) Remove all cation & anion
6) deionised water is useful for high pressures.

* **Disadvantages.**

1. Costly & Chemicals, resin used are expensive.
2. If water contain turbidity, it blocked the pores of resins & output of this process is reduced.

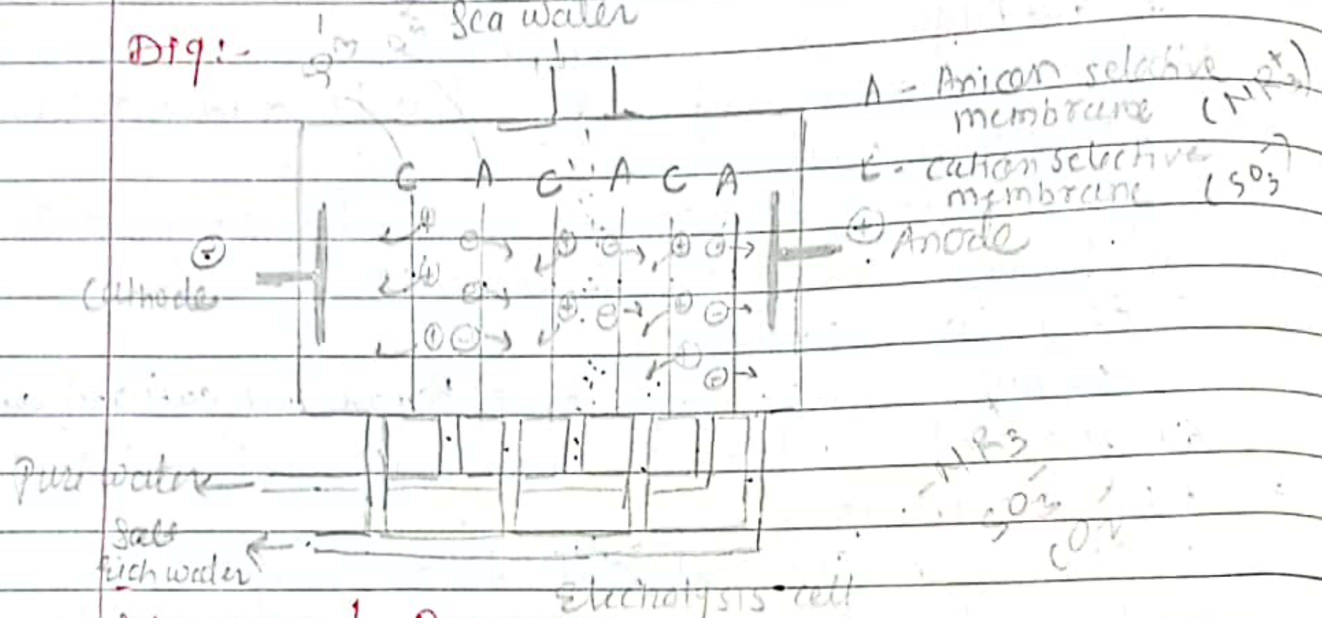


Desalination : Process –

1) Electrodialysis :–

"The process of removing dissolved ionic impurities from water by passing current, electrode & ion selective membrane.

Dig :–



Working / Process :

1) This unit consist two electrodes i.e. Cathode & anode.
2) This chamber consist of selective membrane. Anion selective membrane consist $-NR_3^+$ which repel cation.
3) The cation selective membrane consist of SO_3^-, CO_2^- which repel anion & allow only cation.
4. After applying electric current, emf anion moves toward positively charged electrode & cation toward –ve charged electrode. (cathode)
5. The net result of this migration is depletion of ion in one compartment & conc. of ions in other compartment. So alternate pure & salt water obtained.

* Advantages :–

1. Operation cost is less
2) Equipment is compact

* Limitations:-

1) does not remove colloidal & dissolved organic matters.

2) membrane reqrd maintenance.

* Applications:-

1) Removal of salts from sea water to get pure water.

2) Recovery of salts from salt concentrated water.

3) Removal of ionic pollutant (ions, toxic salts) from treated industrial waste.

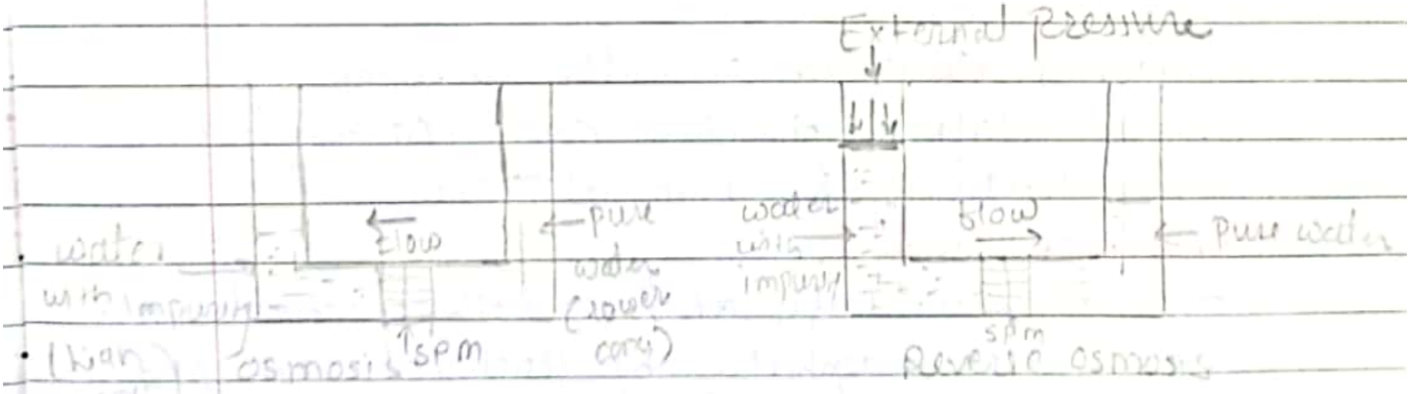
4) Desalination of contaminated industrial water for reuse.

2) Reverse Osmosis:-

Osmosis - flow of soln th' SPM from low conc to higher conc.

Solvent flow from higher concentration solution to lower concentration solution th' semipermeable membrane, by applying external pressure (hydraulic) higher than osmotic pressure of higher concentration sol?

Fig:-



Working:-

1) In this process external pressure is apply on (15-40 kg/cm²) on higher conc. solution slightly higher than osmotic pressure.

2

2 the flow of water takes place in reverse direction from higher to lower conc. i.e from sea water (impure water) to treated pure water, through semipermeable membrane.
3 Treated water comes out from the bottom outlet.
4 Semipermeable membrane is based on thin films of cellulose acetate, polymethyl acrylate, polyamide polymers are used.

* Advantages:-
1 RO remove all ionic, colloidal, nonionic impurity from water. as well as bacteria & viruses.
2) Simple to operate
3) Low cost process.
4) Required less time.
5) obtained drinking mineral water by special SPM membrane by allowing limited quantity of salts.

Application of RO:-
1) Household drinking water purification.
2) Desalination of sea water.
3) Purification of industrial waste water.

* Internal treatment of water in boiler.
method - calgon, phosphate cond.n, carbonate cond.n EDTA treatment, sod. Aluminate etc. added to boiler

phosphate cond.n :-
This method is useful at high pressure. Hard scale converted to phosphates by treatment with Na phosphate & are removed as sludge by blow down operation.

Chemicals in Scale Na.phosphate sludge.

the scale contains $CaCO_3$, $Mg(OH)_2$, $CaSO_4$, SiO_2 fused together as coating

Reac^n :-

Sod. orthophosphates attack on scales to convert them to sludge.

$3CaCO_3 + 2NaH_2PO_4 \longrightarrow Ca_3(PO_4)_2 + Na_2CO_3 + 2H_2O + 2CO_2\uparrow$

$3Mg(OH)_2 + 2Na_3PO_4 \longrightarrow Mg_3(PO_4)_2 + 6NaOH$

If boiler water produce acidic cond^n then alkaline orthophosphate used. If H_2O It produce alkaline cond^n then acidic NaH_2PO_4 is used.

* Phosphate cond^n treatment convert.

1) inside scale to sludge.
2) Hard water to soft water.

Calgon. $Na_2[Na_4(PO_3)_6]$ or $Na_6P_6O_{18}$

name - Sodium hexa phosphate

$Na_2[Na_4(PO_3)_6] \rightleftharpoons 2Na^+ + [Na_4(PO_3)_6]^{2-}$

Principle - $2CaSO_4 + Na_2(Na_4(PO_3)_6] \rightleftharpoons [Ca_2(PO_3)_6]^{2-} + 4Na^+ + Na_2SO_4$

$2Mg^{+2} + [Na_4(PO_3)_6]^{2-} \longrightarrow [Mg_2(PO_3)_6]^{2-} + 4Na^+$

soluble complex

Calgon is better than phosphate cond^n (all pH as appl^n)